Reference Event 1: The Denial of Large-Transformer Replacement in Ukraine, 2022 to 2023

Characterization Against the Critical Manufacturing Sector Design Basis

DBA-CM-RE-1 — Reference Event Instrument, DBA-CM Series

Lineage: Issued under the Critical Manufacturing Sector Framework v0.4; DBA-MA Parent Framework; conferred on the proposal of Long-Lead Critical Components (DBA-CM-LL-FC1)

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose, Scope and Position

1.1 Purpose

This instrument reads the denial of large-transformer replacement in Ukraine since 2022 back through DBA-CM-LL-FC1 (*Long-Lead Critical Components*), whose governing military-action scenario, the asset and its replacement struck together, it characterizes (§6.3), at the class's first reference facility, the large-power-transformer plant (§1.3), and at the replacement path the plant serves (§4). It asks whether the class's design basis anticipated what happened, and returns what it finds.

1.2 Position and Scope

A reference event instrument, in the series convention, reads a real event backward through the design basis, adds no requirements of its own, and returns findings and recommendations for disposition. This one is issued under the Critical Manufacturing Sector Framework v0.4, inherits the DBA-MA parent framework, and was conferred by the framework's §9.1 on the author's decision of 14 October 2026, on the proposal of *Long-Lead Critical Components*, as a continuing condition for this instrument to bound. It is the series' first reference event instrument. Section references of the bare form §N are to that class at v0.1, the version this instrument read.

The condition is bounded as follows. It begins on 24 February 2022, with the full-scale invasion and the closure of the country's largest transformer plant. Its second phase begins on 10 October 2022, the first day of the campaign of strikes on the power system that the International Criminal Court's warrants cover. The reading closes in October 2023, with the delivery of the first high-voltage autotransformers bought with donor funds and the published plan for one hundred new transformers. Later strikes are recorded only as showing the condition continued. The condition continues, and the record is open.

No individual is named; officials are named by role, and the plant, the transmission operator, the court, the United Nations bodies and the governments as institutions. The strikes are attributed to the Russian armed forces by the court's warrants and by Ukraine and its partners; the Russian president stated on 10 October 2022 that Russia had delivered strikes on Ukraine's energy, communications and military command facilities in response to what he called terrorist attacks. Each account is stated as its author's, and this instrument makes no finding of attribution and none of fault.

The plant and the replacement path lie within this class; the installed transformers and their protection are the energy series'. The Foundational Definitions are cited by designation, FD-REG holding their versions: FD-DBL (*Design Basis Layers*) bears on Section 5.

2. Sources and Attribution

2.1 Sources and Their Limit

The sources are: Transformer Magazine's report of 7 March 2022 on the plant's closure; Reuters' report of 7 November 2022 on the government's taking control of the company under martial law; the Atlantic Council's UkraineAlert of 13 January 2023, for the transmission operator's statement of its need; the United Nations Development Programme's energy damage assessment as released on 20 June 2023 and its release of 5 October 2023 on the delivery with Japan; The Economist's report of 2 October 2023 on the plan for one hundred transformers; the International Criminal Court's announcement of 5 March 2024; and the Russian president's statement of 10 October 2022 as reported. The conferral note asked that the plant's and the operator's own statements on the plant's status since 2022 be sought. None was found in a form that could be read: the plant's own announcements and the trade-press reports of its later shipments could not be retrieved. The plant's status after March 2022 rests on the sources above and is marked open.

2.2 Where the Sources Differ

The plant was reported closed in March 2022; in November 2022 the government took control of the company; in October 2023 The Economist reported that few of the transformers ordered from Ukrainian makers had been built because of attacks on Ukrainian factories. No source read states when, or whether, the plant resumed full production. Each is stated with its date.

3. The Event

3.1 The Plant

Zaporozhtransformator, in Zaporizhzhia, was described in March 2022 as the largest transformer plant in Europe, with an annual capacity of 60,000 MVA and about 3,500 employees, and was the transmission operator's domestic source of high-voltage autotransformers. It was reported on 7 March 2022 to have been forced to close by the war. On 6 November 2022 the government took control of the company, with others, under martial law, as necessary in the conditions of war (Reuters).

3.2 The Campaign on the Installed Base

From 10 October 2022 the power system was struck in repeated waves. The Russian president stated that day that the strikes were on energy, communications and military command facilities, using high-precision long-range weapons. The International Criminal Court found reasonable grounds to believe that two Russian commanders bear responsibility for a campaign of strikes on electric power plants and substations from at least 10 October 2022 to at least 9 March 2023, and issued warrants on 5 March 2024.

3.3 The Transformers Lost

As of 11 January 2023 the transmission operator stated that the country needed fifty-nine large autotransformers; each cost about $2 million and weighed hundreds of tons, could take up to two years to buy and receive, and was unlikely to fit available cargo aircraft, so that delivery would be by sea over several months (Atlantic Council). The United Nations Development Programme's damage assessment found that 42 of the 94 vital high-voltage transformers in government-controlled territory, about 45 percent, had been damaged or destroyed between January and April 2023, and that more than half of them had endured repeated attacks, which blocked attempts to repair them. The energy minister said in October 2023 that autotransformers had been key targets in the previous winter's attacks.

3.4 The Replacement Path

In October 2023 the United Nations Development Programme, funded by Japan, delivered two high-power autotransformers, which the operator's chief executive described as the first new high-voltage autotransformers bought with international donor funds, delivered faster than such procurement usually takes. The same month The Economist reported that about one hundred new high-voltage transformers had been ordered, half to be made in Ukraine and half abroad, the foreign units to be held in Poland and Romania until needed, and that few domestic units had been built because of attacks on Ukrainian factories.

4. Technical Decomposition

The condition decomposes into initiating elements (I-1 to I-3) and effect elements (E-1 to E-5), as Table 4-1 sets out. Composed, they are the class's governing scenario (§6.3) at the large-power-transformer plant and its replacement path, in Chain C, with Chain R opened by the loss of the domestic plant.

Table 4-1. Event elements and their mapping. Section references are to *Long-Lead Critical Components* v0.1.

No.	What happened	Class provision	Section
I-1	Largest domestic plant closed by the war; factories attacked	Qualified source count (§3.3); Chain R (§3.2)	3.1, 3.4
I-2	Campaign on the power system from October 2022	Asset and replacement struck together (§6.3); Chain C	3.2
I-3	Autotransformers named as key targets	Precision (§6.2)	3.3
E-1	Fifty-nine autotransformers needed; 42 of 94 vital transformers damaged or destroyed	Reserve coverage (§3.3, §4.4)	3.3
E-2	Repeat strikes block repair of more than half	Repair; first residual (§6.8)	3.3
E-3	Up to two years to buy; delivery by sea over months	Build and queue time; delivery (§3.3, §4.5)	3.3
E-4	First donor-funded autotransformers in October 2023	Emergency order path (§4.2)	3.4

E-5	Foreign units to be held in Poland and Romania	Reserve beyond the campaign's reach (§6.3)	3.4

5. Tier Determination

5.1 The Four Characteristics

The characteristics are read against the campaign on the installed base and on the plants that would replace it, in the forms *Long-Lead Critical Components* §6.2 states; the author ruled on 14 October 2026 that they may be read against a campaign on what a class's facilities depend on. Intentionality is present: the strikes on energy facilities were deliberate on the Russian president's own statement and on the court's finding. Persistence is present: the strikes came in repeated waves through the winter of 2022 to 2023 and continued after it. Precision is present: autotransformers were key targets, on the energy minister's account, and they are the component with the longest replacement and the fewest sources. Denial is present: more than half of the damaged transformers were struck again, blocking repair, and the domestic plants that would have built replacements were closed or attacked.

5.2 Determination

The event meets the threshold for the military-action extension. All four characteristics are established on the record, as FD-DBL (*Design Basis Layers*) §6.4 requires under the author's rulings of 10, 13 and 14 October 2026, and the actor is a state's armed forces on the court's finding. It is a military-action event of the class's governing scenario (§6.3), in Chain C, at the reach of a country and a duration of years. That is the tier determination returned for the DBA Reference Event Register. Energy is proposed as an impacted sector, as the sector whose recovery the event bounds.

6. Characterization Against the Design Basis

6.1 §6.3, §2.4 and §4.4 The Class's Account of the Event

Held, with a gap. The account holds in every figure checked: the plant's closure as reported in March 2022, its capacity, the fifty-nine autotransformers, the transport constraint, the October 2023 delivery and the plan for one hundred transformers held abroad. Two points are qualified and go to the class as Erratum 1 of Project 11's note of 14 October 2026 (Critical Manufacturing batch): the plant's status after March 2022 should be stated as open, with the government's taking control of the company in November 2022 and The Economist's report of attacks on factories, since neither the plant's nor the operator's own statements could be read; and §6.3 should add the damage assessment's 42 of 94 transformers and its finding that repeat strikes blocked repair, which is the denial the class postulates.

6.2 §6.3 The Asset and Its Replacement Struck Together

Held. The installed autotransformers were struck in a campaign while the domestic plant that built them was closed and other factories attacked; recovery came to depend on foreign plants, foreign spares and foreign routes. The event is the class's governing scenario carried out.

6.3 §6.8 and §3.3 Repair Under Repeat Strikes

Held, with a gap. The class names the repair of damaged units, where possible, as part of the response beyond the design campaign, but its five parameters count build, queue, sources, surge and reserve, and none counts the capacity to repair. The campaign aimed at that capacity: more than half of the damaged transformers were struck again, blocking repair. Finding F-1.

6.4 §4.2 The Emergency Order Path

Held, with a gap. The class requires an emergency order to be placed ahead of the queue by a planned authority. The event shows the binding constraint was not the queue alone but a buyer with funds and procurement authority: the first new autotransformers bought with donor funds arrived in October 2023, nine months after the operator stated its need, through an international agency acting as buyer. Finding F-2.

6.5 §4.4 and §6.3 The Reserve Beyond the Campaign's Reach

Held. The plan to hold foreign units in Poland and Romania until needed is the reserve the class requires, held where the campaign cannot reach it.

6.6 §4.5 Test, Transport and Installation

Held. The largest units could not go by air and would go by sea over months, as the class records; the delivery time was part of the recovery time.

6.7 §6.4 Denial by Decision

Not exercised. No supplying plant or material was barred from Ukraine by a decision in the sources read.

6.8 §7.3 and §7.4 Allocation Under a Campaign

Not exercised. The sources read do not describe how units were allocated among waiting substations, or any component within the scope of *Layer Decomposition and Determinism* §4.3 proposing it.

6.9 Summary of Design Basis Performance

Of the eight readings, three Held, with a gap, the class's account among them; three Held; and two were Not exercised. Nothing was Missed. The class's governing scenario was written with

this event as its case, and the event is the scenario carried out. What the event adds is that the campaign aimed at repair as well as replacement, and that the emergency order path needs a buyer with funds before it needs a place in the queue.

7. Findings

Section references are to *Long-Lead Critical Components* v0.1 unless another instrument is named.

7.1 F-1 — Repair Capacity Is Part of the Replacement Path

The damage assessment found that 42 of 94 vital high-voltage transformers had been damaged or destroyed in four months of 2023, and that more than half had been struck repeatedly, which blocked repair (Section 3.3). The finding confirms §6.8's naming of repair as part of the response, and is that the class should state repair capacity as a sixth parameter of §3.3: the capacity of the plants, service crews, parts and test equipment to return a damaged unit to service in place, and the time it takes; that a first-tier plant's design basis states the repair it supports for the units it built; and that, in the military-action domain, repair capacity is credited only where it and the unit under repair can be protected from a repeat strike, which is the energy series' to state for the installed unit.

Affected: §3.3, §6.3, §6.8. Disposition: R-1; noted to the energy series in Project 8 for the protection of installed units.

7.2 F-2 — The Emergency Order Path Includes the Buyer and the Funds

The operator stated in January 2023 that it needed fifty-nine autotransformers; the first new autotransformers bought with donor funds arrived in October 2023, through an international agency acting as buyer, which the operator credited with moving faster than such procurement usually takes (Sections 3.3 and 3.4). The finding confirms §4.2's requirement that the emergency order path be planned before it is needed, and is that the class should state that the planned path for a first-tier specification names, in advance, the buyer that would place an emergency order, the source of its funds, the procurement rules it would follow and the qualified plants beyond the campaign's reach with which it would place the order, so that the queue time of §3.3 begins on the day of the loss.

Affected: §4.2, §4.4, §6.3. Disposition: R-2; noted to the coordination class, CorpCo.

8. Recommendation Register

Priority is on the series' one-to-three scale, 1 the most consequential. R-1 and R-2 dispose to *Long-Lead Critical Components*.

No.	Title	Affected	Recommendation	Pri.

R-1	Count repair capacity in the replacement path	§3.3, §6.3, §6.8	State repair capacity and time as a sixth parameter; plants state the repair they support; credit repair under military action only where protected from repeat strikes. Noted to Project 8.	1
R-2	Name the buyer and the funds in the emergency path	§4.2, §4.4, §6.3	Name in advance the buyer, its funds, its procurement rules and the qualified plants beyond the campaign's reach. Noted to CorpCo.	1

9. Open Items

OI-1: the register row. Project 0 is asked to enter the tier determination of Section 5.2 and the impacted sector. OI-2: the class has not taken in this instrument and conforms at its next revision, with Erratum 1. OI-3: the plant's status after March 2022 is open; the instrument is revised when its own or the operator's statements can be read.

10. Standing of This Document

10.1 What This Instrument Establishes

It modifies neither class nor framework, and makes no finding of attribution and none of fault.

10.2 Relationship to the Class's Use Case

The class's use case traces §6.3. This event is that scenario carried out, so the use case may run on its figures: the country's largest domestic plant lost at the start; 42 of 94 vital transformers damaged or destroyed in four months, more than half struck again; fifty-nine autotransformers needed, each up to two years to buy and months by sea; the first donor-funded units nine months after the need was stated; and a reserve planned abroad. F-1 and F-2 bear most on it.

10.3 Relationship to Other Instruments

Ukraine's Payment System Through the War (DBA-FIN-RE-3) and *Air-Raid Alerting in Ukraine* (DBA-ESS-RE-9) read the same campaign through other sectors. *The Shortage of Transformers in the United States* (DBA-CM-RE-2) reads the replacement path stretched by demand alone. The concrete shelters the operator has since built over many of its transformers, as its former chief executive described them, bear on the protection of installed units, which is the energy series'.

References

Transformer Magazine, *Largest transformer factory in Europe closed due to war*, 7 March 2022.
Reuters, *Ukraine seizes stakes in strategic companies under wartime laws*, 7 November 2022.
Atlantic Council, UkraineAlert, *Ukraine struggles to repair power grid as Russian airstrikes*

continue, 13 January 2023. United Nations Development Programme, *UNDP energy damage assessment for Ukraine reveals continued vulnerabilities*, 20 June 2023; release on the delivery of high-power autotransformers with Japan, 5 October 2023. The Economist, report of 2 October 2023 on Ukraine's transformer orders. International Criminal Court, announcement of 5 March 2024. Novaya Gazeta Europe, report of the Russian president's statement of 10 October 2022. Interfax-Ukraine, report of 5 November 2025 on transformer shelters. The DBA instruments named in the text, with the Foundational Definitions and FD-REG, and the DBA-MA Parent Framework.

Revision History

Version 0.1, 14 October 2026: issued in Project 11 under the Critical Manufacturing Sector Framework v0.4, on the conferral of the framework's §9.1 of 14 October 2026, read against *Long-Lead Critical Components* v0.1. Written new under the Reference Event Compression Standard v0.3, the series' first reference event instrument, bounding a continuing condition from 24 February 2022 to October 2023. Tier determination at Section 5.2, under FD-DBL §6.4 as ruled on 10, 13 and 14 October 2026; F-1 and F-2, R-1 and R-2. The class's account holds, with two qualifications going to the class as Erratum 1 of Project 11's note of 14 October 2026 (Critical Manufacturing batch).